BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 039 · Utility Software

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. Why does an operating system use a device driver?

Q2. What does anti-malware detect?
A. Every future threat
B. All hardware faults
C. Malicious software
D. All weak passwords

Q3. Why use different user access levels?

Q4. A character code uses exactly 7 bits. How many different characters can it represent? Can it encode 129 different characters without more bits?

I Do · Read new knowledge

Key vocabulary

Utility software — Programs performing additional maintenance or protection tasks.

Encryption — Converts plaintext into ciphertext using an algorithm and key.

Decryption — Converts encrypted data back to readable form with the required key.

Compression — Reduces how much storage data requires.

Defragmentation — Rearranges fragmented file parts on a magnetic hard disk.

Fragmentation — Parts of a file stored in separated locations on disk.

Magnetic hard disk — Secondary storage device using rotating magnetic platters.

New learning

Utilities versus the OS: Utility software performs additional housekeeping, maintenance and protection tasks which might not be performed directly by the operating system.

Encryption utility: Encryption changes readable plaintext into ciphertext using an algorithm/key, protecting data if it is accessed without authorisation. A suitable key is needed to decrypt it.

Compression utility: Compression reduces the storage space needed for data. Lossless allows exact reconstruction; lossy removes information and is unsuitable when exact recovery is needed.

Defragmentation: On a magnetic hard disk, fragments of a file may be scattered. Defragmentation puts file parts together so the read head needs fewer movements. SSDs should not be routinely defragmented for this purpose.

Worked example

A school library's magnetic hard disk has become fragmented. Explain how defragmentation can improve file access. [4 marks]

1. Identify the storage medium: a magnetic hard disk.

2. Parts of files may be scattered in separate locations on disk [1].

3. Defragmentation rearranges these parts into adjacent blocks [1], reducing movement of the disk read/write head [1].

4. Less movement can reduce time accessing the file [1]. Do not suggest this improves SSD read-head movement.

We Do · Practise with guidance

// BLACK PEN Answer all questions first. //✓ GREEN PEN Then: tick and rewrite mistakes.

Q3. Compression reduces file size [1] to save storage or transfer time [1]. Encryption converts data to unreadable ciphertext [1], protecting content from unauthorised access [1]. Defragmentation reorganises fragmented HDD files [1] and can reduce mechanical disk access time [1].

Exit Ticket · Model answer

True

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. It enables communication with and control of a particular peripheral.

Q2. C — Malicious software

Q3. Give users only the access rights they need. This limits access without permission and possible damage.

Q4. (a) 2^7 = 128; no, one more character requires more distinct codes.

I Do / We Do pages · Guided practice

Q1. Encryption turns readable data into ciphertext [1] using an algorithm/key [1]. If the device is lost or stolen [1], the records cannot readily be understood without the correct decryption key [1].

Q2. Compression software reduces file size [1], saving storage or transfer time [1]. Lossy compression discards some data [1], so the original sound cannot be perfectly reconstructed [1].

Q3. On an HDD files can be fragmented [1], making the mechanical read/write head move between locations [1]. Defragmentation groups parts and reduces head movement [1]; SSDs have no moving read/write head, so this benefit does not apply [1].

You Do pages · Code First

```
originalKB = 240
compressedKB = 60
print(originalKB - compressedKB)
```

Expected result: 180

You Do pages · Independent questions

Q1. Defragmentation [1].

Q2. Encryption transforms readable data into ciphertext [1], preventing straightforward reading [1]. A decryption key is needed [1], so losing the drive does not automatically expose readable records [1].

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Guided 1

A business stores confidential customer records on portable media. Explain why encryption software is suitable. [4 marks]

Worked method: Encryption turns readable data into ciphertext [1] using an algorithm/key [1]. If the device is lost or stolen [1], the records cannot readily be understood without the correct decryption key [1].

Guided 2

A podcaster wants to reduce the size of large audio files and accepts some lost quality. Explain a suitable utility and one drawback. [4 marks]

Worked method: Compression software reduces file size [1], saving storage or transfer time [1]. Lossy compression discards some data [1], so the original sound cannot be perfectly reconstructed [1].

Guided 3

A school has an SSD and a magnetic hard disk. Explain why defragmentation may help the HDD but is not normally useful for the SSD. [4 marks]

Worked method: On an HDD files can be fragmented [1], making the mechanical read/write head move between locations [1]. Defragmentation groups parts and reduces head movement [1]; SSDs have no moving read/write head, so this benefit does not apply [1].

BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Write the line that prints storage saved in KB.

```
originalKB = 240
compressedKB = 60
```

write

Question 1

Name the utility that rearranges file blocks on a magnetic hard disk. [1 mark] Source: Original OCR J277-style teaching question; not an official past-paper question.

Question 2

A doctor encrypts confidential files on a portable drive. Explain TWO reasons this helps keep information private. [4 marks] Source: Original OCR J277-style teaching question; not an official past-paper question.

Question 3

A school is choosing utility software. Explain when compression, encryption and defragmentation may each be useful. [6 marks] Source: Original OCR J277-style teaching question; not an official past-paper question.

RHS COMPUTING / OCR J277 / 1.5.2 LESSON 039 · DRAFT

BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

Exit Ticket · Answer before checking

Defragmentation is mainly intended to reduce mechanical read-head movement on fragmented magnetic hard disks.